

PEIYU LIU

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SUMMARY

- Ph.D. candidate in **Biostatistics** at University of Florida with **7+ years'** experience in statistical modeling, differential equations, and prediction modeling in clinical and simulation-based research.
- Proficient in multiple coding languages including **R (tidyverse, Shiny)**, **SAS**, **Julia**, **Python**, **Matlab**, **Maple** etc.
- Specialized in Bayesian statistics (**MH**, **INLA**, **MAP estimation**, **prior sensitivity analysis**), Bayesian hierarchical modelling (**BYM2** with **penalized priors**, mixed-effect and **latent-variable models**), **missing-data methods** (**data augmentation**, **missing-not-at-random sensitivity**), **CDISC** (**ADaM**, **SDTM**, **TLG**), clinical trial design (**estimand**, **sample size**), and **optimize dose selection** (**PK-BOIN12**, **ER-CUS**, **STEIN12**).

EXPERIENCES

Biostatistics Intern

Remote, U.S.

Hematology & Cell Therapy Biostatistic | Bristol Myers Squibb

May 2026 - Aug 2026

- Designed a **clinical utility score (CUS)** Rshiny for hematology assets, **weighting efficacy, safety, and tolerability endpoints** into a single comparable measure, incorporated **clinical and HEOR feedback** into weighting scheme.
- Extended the CUS framework from binary endpoints to **mixed endpoint types** — continuous and time-to-event, making the RShiny app score assets on **response, tumor shrinkage, and survival** within one utility scale.
- Built a tool comparing **FDA Project Optimus**—aligned dose-optimization designs (**PK-BOIN12**, **STEIN**) for phase2 trials, running the **adaptive dose-finding** and returning an **optimal biological dose** without hand-coding.
- Created a **CDISC traceability AI agent** resolving an **analysis value** back through **ADaM**, **SDTM**, and raw layers, turning a multi-hour manual lookup into a single query.

Biostatistician

Gainesville, U.S.

Biostatistics Core | Emerging Pathogens Institute | University of Florida

May 2025 - Present

- Led biostatistical support and **PK/PD modeling** for **multiple faculty research projects**, designing analyses, optimizing methodologies, **delivering actionable statistical insights across infectious disease and vaccine studies**.
- Modeled measles outbreak risks by **compartmental models** parameterized by **10 years** of county-level vaccination data, **incorporating regional covariates** to evaluate outbreak size and probability across coverage scenarios.
- Engineered and released the **open-source R package *vaxineR***, which **standardized** outbreak risk modeling workflows—including R_e estimation, vaccination scenario simulation, and county-level visualization—**enhancing reproducibility across studies**.
- **Generated high-impact visualizations and quantitative reports** that were featured in *Science News*, elevating public awareness of vaccination coverage decline and its epidemiological consequences.
- Designed and executed **statistical analyses of RSV severity using EHR data**, applying **Cox proportional hazards and logistic regression models** to evaluate the effects of age, gender, birth history, and comorbidities on length of stay, ICU admission, oxygen use, and ventilation outcomes.
- Developed **comprehensive model evaluation reports** with feature importance rankings, diagnostic visualizations, and interpretive summaries, contributing analytic figures and results to a **manuscript submission to Microorganisms**.
- **Led statistical analyses in a pediatric clinical-genomic RSV study**, integrating **Cox and multivariable regression models** with **viral genotype data** to uncover genotype–phenotype associations and their influence on disease severity.
- **Collaborated with clinicians and virologists** to translate analytical findings into biologically interpretable results, supporting ongoing research in pediatric respiratory infections.

Research Assistant

Gainesville, U.S.

Biostatistics Department | University of Florida

Aug 2023 - Present

Project: Trial Endpoint Optimization for Arboviral Vector-Control Interventions — First author

- Independently proposed and designed a **methodological framework** comparing three **candidate endpoint** strategies (serological biomarker, clinical cases, vector density) for **cluster-randomized** vector-control trials.
- Derived **analytical sample-size formulas** of antibody **Non-homogeneous Poisson process** model and validated them

across **realistic trial design space** (cluster size, follow-up duration, intervention efficacy) through ~10,000 simulated **trial replicates** per scenario.

- Enabled **a priori power calculations** under three **parallel sample-size frameworks**: **Freedman's method (time-to-event)**, two-sample t-test (continuous biomarker), and **negative binomial regression** (count outcomes via MKmisc).
- Delivered **oral presentations** at PHHP days, **communicating** statistical methodology to **mixed audiences** of statisticians, clinicians, and public-health faculties, demonstrated up to 6-order-of-magnitude **sample-size reductions** for the **biomarker endpoints** in low-incidence settings.
- Translated methodology into a deployable interactive [Shiny application](#), enabling clinical and epidemiological collaborators without statistical background to perform endpoint exploration and power analysis in real time.

Project: Practical identifiability and Bayesian inference of seasonal exposure dynamics from cross-sectional serology — First author

- Developed a **Bayesian inference framework** for seasonal exposure dynamics where the exact likelihood requires **intractable trans-dimensional integration** over **unobserved individual-level latent variable histories** — a class of **missing-data problem** with no off-the-shelf solution.
- Derived a computationally tractable **surrogate likelihood** for a mechanistic NHPP model, using the **Campbell–Mecke theorem** for **moment-matching**, reducing trans-dimensional integration into an analytical Gamma observation model.
- Conducted pre-specified **practical identifiability analysis** using **Fisher Information Matrix eigendecomposition** and a **coordinate identifiability metric** across three cross-sectional study designs (repeated, multi-site, interventional).
- Implemented **affine-invariant ensemble MCMC** (emcee) with rigorous **convergence diagnostics** (**autocorrelation function, effective sample size, trace plots**) across 24 independent posterior runs over a **prior sensitivity grid**.
- Designed inference framework with **deliberately distinct data-generating and inference models** to preclude the **inverse-crime** and provide a rigorous test of out-of-model predictive validity.
- Applied framework to three published real cohorts (Elanga–Ndille 2012 Benin; Fustec 2021 Thailand, two sites), recovering interpretable seasonal exposure profiles consistent with independent entomological surveillance.

Project: CHIKV Household Transmission Modelling with MCMC data augmentation and Bayesian hierarchical spatial modelling — First author

- Designed a pre-specified three-model framework (**GEE marginal**, **BYM2 Bayesian hierarchical**, Longini-Koopman mechanistic) sharing a unified 12-dimensional covariate parameterization, enabling direct cross-model comparison of community vs household transmission effects.
- Diagnosed and corrected a normalization error in the **data-augmented MCMC sampler** for the Longini-Koopman household final-size model: derived the binomial-coefficient correction in the **Gibbs full conditional** for **missing-member latent states** from first principles.
- Derived the **observed-data marginal likelihood** integrating over **missing-member latent states** via hypergeometric **sampling weights**.
- Diagnosed **missing-not-at-random patterns** in window-screens covariate (36.5% missing, correlated with income and outcome); pre-specified missing-as-category primary analysis with **multiple-imputation (MICE) sensitivity**.
- Implemented **production-grade MCMC** in **Julia** with multi-threaded 4-chain $\times$ 50,000-iteration sampling and G-pattern caching optimization, achieving $\sim 10\times$ **speedup** over baseline; verified convergence via **Gelman-Rubin $\hat{R} < 1.05$** and **ESS $\geq 1,000$** across all parameters in all chains.

Research Fellow and Lecturer

Math Department | Hunan Institute of Science and Technology

Changsha, China

Jun 2021 - Aug 2023

- Taught **Probability Theory** and **Mathematical Statistics** to 200+ students, 243 class hours each year.
- Developed a **quantitative modeling framework** using a **three-dimensional pulsed differential equation model** to simulate interactions among **tumor cells, effector cells, and helper T cells**, providing insights into **tumor-immune system dynamics** and treatment response.
- Analyzed treatment parameters through **bifurcation and sensitivity analyses** to evaluate how **therapy intensity** and immune activation thresholds **influence** tumor control and intervention efficacy, **supporting optimization of radioimmunotherapy strategies**.
- Completed and harmonized **multinational weather datasets** using spline interpolation to address data missingness,

and led spatial analyses with bivariate choropleth mapping to visualize COVID-19 spread.

- **Applied machine learning techniques**—including **Expectation Maximization (EM)**, **Lasso regression**, and **Generalized Additive Models (GAM)**—to assess associations between weather variables and COVID-19 transmission across 10 countries.

Researcher

Xian, China

School of Mathematics and Statistics | Shaanxi Normal University

Jul 2018 - Jun 2022

- Integrated **HIV viral dynamics** with **pharmacokinetic** and **adherence processes** in a **four-dimensional ODE** system to model viral blips, calibrated model parameters using **MCMC in Stan**.
- Developed a **stochastic two-compartment pharmacokinetic model** with **binomial adherence behavior** to simulate **drug concentration variability** under random **dose omission**.
- Built and calibrated **SEIAR compartmental models** using **Nonlinear Least Squares**, **Bootstrap**, and **MCMC methods**, performing **sensitivity and threshold analyses** to quantitatively evaluate and inform vaccination and non-pharmaceutical intervention (NPI) strategies.
- Developed and maintained the compartmental modeling components for a **data-driven COVID-19 forecasting and visualization platform**, enhancing the accuracy and interpretability of epidemic projections for public health decision-making.

EDUCATION

School of Public Health & Health Professions, University of Florida

Gainesville, U.S.

Ph.D. in Biostatistics, GPA: 3.9/4

Aug 2023 - Present

- **Coursework:** *Biostatistics Methods and Computing, Statistical Machine Learning, Survival Analysis, Causal Inference, Data visualization, Longitudinal Analysis, Analytical methods for Infectious Diseases*

School of Mathematics and Statistics, Shaanxi Normal University

Xian, China

M.S. in Computational Mathematics, top 5%

May 2021

- **Coursework:** *Advanced Biomathematics, Stochastic Process, Stability, Qualitative Theory of Differential Equations*

School of Mathematics and Statistics, Shaanxi Normal University

Xian, China

B.S. in Statistics, top 5%

May 2017

- **Coursework:** *Probability Theory, Statistical Analysis and R, Mathematical Modeling, Practical Regression Analysis, Multivariate Statistical Analysis, Statistical Computing, Time Series Analysis*

APPLICATIONS/PACKAGES

R Package

- Peiyu Liu, Matt Hitchings, Ira Longini. Package “vaxineR” Statistical toolkit for vaccine-coverage analysis and outbreak-risk modeling. <https://cran.rproject.org/web/packages/vaxineR/>

Clinical Data Tools

- Clinical Data Trace — CDISC lineage tracing from an analysis value back through ADaM, SDTM, and raw layers: <https://agent-cdisc-datasets-trace.streamlit.app/>
- Sample Size Suite — CDISC-aligned dataset checks and sample size calculation across time-to-event, continuous, and count endpoints: <https://peiyuliu.shinyapps.io/clinical-data-ssize-suite/>

R-Shiny Apps:

- Oncology Dose Optimization (Clinical Utility Score framework, exposure-response modeling, PK-BOIN dose finding, STEIN): <https://peiyuliu.shinyapps.io/oncology-cus-pkboin-v1/>
- Dengue Clinical Trial (serological vs. clinical endpoints): <https://peiyuliu.shinyapps.io/anpp-trial-comparison/>
- US Weather Analyses (ggplot2 visualization, K-means clustering, Machine Learning forecasting): <https://peiyuliu.shinyapps.io/us-weather-analysis/>
- SIR Model Estimation (Least Squares, Bootstrap, MCMC): <https://peiyul.shinyapps.io/sir-estimation-comparison/>

GOOGLE SCHOLAR

- Google Scholar Available at: <https://scholar.google.com/citations?user=O6KFoOUAAAJ&hl=en&oi=ao>